Innovarsity Robotics Research Internship (ROS2 Track)

15-Day Intensive Training & Research Plan

Role: Robotics Research Intern (ROS2)
Organization: Innovarsity

Internship Objective

The objective of this internship is to prepare the research intern to:

- Develop a strong foundation in Ubuntu Linux and robotics programming.
- Understand ROS2 architecture and ecosystem from first principles.
- Install, configure, troubleshoot, and maintain ROS2 development environments.
- Develop ROS2 packages, nodes, services, actions, and launch systems.
- Interface with robotic hardware.
- Build and test applications using a **4-DoF Dobot Magician** as a Proof of Concept.
- Progress towards contributing to the **6-DoF Taurean Surgical Robot Software Controller Stack**, including motion control, hardware abstraction, and modular software architecture.

Expected Deliverables after 15 Days

The intern should be able to independently:

✅ Install Ubuntu and ROS2 from scratch

✅ Configure Linux environment

✅ Troubleshoot ROS2 installation

✅ Create ROS2 workspaces

✅ Build custom packages

✅ Publish and subscribe topics

✅ Develop Services & Actions

✅ Launch complete robot systems

✅ Understand TF Tree

✅ Interface robot hardware

✅ Operate Dobot using ROS2

✅ Understand robot kinematics

✅ Understand MoveIt2 basics

✅ Contribute to Taurean Surgical Robot software

Week 1 — Ubuntu + ROS2 Foundations

Day 1

Ubuntu & Linux Fundamentals

Topics

- Ubuntu Installation
- Dual Boot / Virtual Machine
- File System
- Terminal Commands
- File Permissions
- SSH
- Nano & Vim
- Package Manager (APT)

Hands-on

- Install Ubuntu
- Install VS Code
- Install Git

- Configure Terminal

Assignment

Build a Linux Cheat Sheet.

Day 2

Linux for Robotics

Topics

- Processes
- Services
- Environment Variables
- Bash Scripts
- Cron
- Networking
- IP Address
- SSH between systems

Hands-on

- Write shell scripts
- Create automation scripts
- Configure static IP

Assignment

Create an Ubuntu setup automation script.

Day 3

Python for Robotics

Topics

- Python Review
- Virtual Environments

- OOP
- File Handling
- Threads
- Exception Handling

Hands-on

- Sensor simulator
- Motor simulator

Assignment

Create a console robot simulator.

Day 4

ROS2 Installation

Topics

- ROS2 Overview
- DDS
- ROS2 Architecture
- ROS Distributions
- Environment Setup

Hands-on

- Install ROS2
- Install Colcon
- Create Workspace

Assignment

Successfully build the first workspace.

Day 5

ROS2 Core Concepts

Topics

- Nodes
- Topics
- Messages
- Publishers
- Subscribers
- CLI Tools

Hands-on

Create

- Talker
- Listener
- Custom Publisher

Assignment

Develop a multi-node communication project.

Week 2 — ROS2 Development

Day 6

ROS2 Packages

Topics

- Package Structure
- CMake
- Python Packages
- Dependencies
- Colcon

Hands-on

Create multiple packages

Assignment

Organize a robotics workspace.

Day 7

Services & Parameters

Topics

- Services
- Parameters
- Configuration
- YAML Files

Hands-on

Robot Parameter Server

Assignment

Create configurable robot parameters.

Day 8

Actions

Topics

- Actions
- Goal
- Feedback
- Result

Hands-on

Robot movement simulator

Assignment

Move robot using Actions.

Day 9

Launch Files

Topics

- Launch Files
- Namespaces
- Multiple Nodes
- Logging

Hands-on

Launch complete robot application.

Assignment

One-click robot startup.

Day 10

TF2 & Robot Frames

Topics

- Coordinate Frames
- TF Tree
- Static Transform
- Dynamic Transform

Hands-on

Visualize robot frames

Assignment

Create TF tree for 4-DoF robot.

Week 3 — Robot Integration

Day 11

URDF & Robot Description

Topics

- Robot Links
- Joints
- Visual Meshes
- Collision
- Inertia

Hands-on

Build robot model.

Assignment

Create URDF for 4-DoF Robot.

Day 12

RViz + Gazebo + MoveIt2

Topics

- RViz
- Gazebo
- MoveIt2
- Motion Planning

Hands-on

Visualize robot.

Assignment

Move virtual robot.

Day 13

Dobot Magician ROS2 Integration

Topics

- Driver Installation
- Serial Communication
- Joint States
- Motion Commands

Hands-on

Control

- Joint Motion
- End Effector
- Cartesian Motion

Assignment

Pick-and-place demonstration.

Day 14

6-DoF Surgical Robot Architecture

Topics

Taurean Software Stack

- Robot Layers
- Hardware Interface
- Controller Manager
- Safety Layer
- Motion Planner
- Inverse Kinematics
- Forward Kinematics

- State Machine
- Diagnostics

Hands-on

Study existing software architecture.

Assignment

Design software module architecture.

Day 15

Final Research Project

Objective

Develop an integrated ROS2 software package.

Deliverables

- Robot Package
- Launch File
- TF Tree
- URDF
- Motion Controller
- Documentation
- Git Repository
- Final Presentation

Demonstration

- ROS2 installation from scratch
- Robot simulation
- Dobot control
- Motion planning
- Controller explanation
- Proposed architecture for the Taurean Surgical Robot controller stack

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Parallel Learning Activities (Daily)

Each day should also include:

Activity	Duration
Ubuntu Practice	30 min
ROS2 CLI Practice	30 min
Git & GitHub	20 min
Technical Documentation	20 min
Robotics Research Paper Reading	30 min
Code Review	20 min
Lab Notebook / Daily Log	15 min

Expected Final Software Stack Understanding

By the end of the internship, the intern should understand the layered architecture below:

```
Application Layer
│
├── Surgical Procedure Applications
├── Teleoperation Interface
├── User Dashboard
│
Motion Planning Layer
│
├── MoveIt2
├── Trajectory Planning
├── Cartesian Planner
│
Control Layer
```

```
|
├── Joint Controller
├── Velocity Controller
├── Position Controller
├── PID Loops
|
Robot Abstraction Layer
|
├── Hardware Interface
├── Sensor Interface
├── Encoder Interface
├── Motor Driver Interface
|
ROS2 Middleware
|
├── Topics
├── Services
├── Actions
├── TF2
├── Parameters
|
Linux Ubuntu
|
Hardware
```

Final Evaluation Rubric

Evaluation Area	Weightage
Ubuntu & Linux Proficiency	10%
ROS2 Fundamentals	20%
Package Development	15%
Topics, Services & Actions	15%
URDF, TF2 & RViz	10%

Dobot Magician Integration	15%
6-DoF Controller Stack Design	10%
Documentation, Git & Presentation	5%